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Prof. Kenneth M. Merz, Jr., Editor-in-Chief  
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Dear Prof. Merz,

We submit our manuscript **“Target breadth and mutation resilience in African-natural-product-inspired anti-malarial chemotypes: a computational analysis”** for consideration as an Article in *Journal of Chemical Information and Modeling*.

This work addresses a critical methodological challenge in computational antimalarial drug discovery: evaluating target breadth and mutation resilience as distinct, complementary properties when prioritizing candidates from African natural-product-inspired chemical space, while maintaining clear boundaries between computational evidence and biological activity. We analyze a 17-member polypharmacology-oriented cohort across four mechanistically diverse antimalarial targets (PfDHFR, PfCRT, PfClpP, and PfATP4) using target-specific structural anchors and a per-target resistance-resilience scoring (RRS) framework. The workflow integrates three evidence layers: chemical-space expansion (65 856 hybrid library molecules  $\rightarrow$  19 913 synthesizable candidates), target-anchored AutoDock Vina docking generating a complete  $17 \times 4$  scoring matrix (68/68 geometric quality pass rate), and per-target RRS calculated separately for PfDHFR (N51I, C59R, S108N, I164L) and PfCRT (K76T, K76A) mutant panels, with PfCRT re-docked against a corrected 3D7-like wild-type receptor under a cavity-anchored grid and a pipeline-null control demonstrating that the corrected PfCRT channel carries no detectable K76T/K76A signal for this panel. The cohort yielded seven A\*, four B, and six C RRS profiles (range 73.2–115.6); after multiplicity correction, PNS–RRS ( $\rho = -0.714$ ), RRS–wild-type score ( $\rho = 0.691$ ), and  $N_{\text{fav}}$ –RRS ( $\rho = 0.714$ ) were significant, while ACSI–RRS was not ( $\rho = -0.190$ ).

**Key enhancements** (comprehensive validation, physicochemical characterization, and methodological documentation):

1. **Validation documentation:** three complementary approaches: (a) external benchmark (DEKOIS 2.0, PfDHFR ROC-AUC 0.45, reported honestly as near-chance), (b) enrichment validation (MMV Malaria Box, ROC-AUC 0.924–1.000), and (c) redocking validation (four of five reference ligands reproduced with RMSD  $< 2.0$  Å; the methotrexate exception reflects NADPH-adjacent grid placement). Metrics in Supporting Information Section S12.
2. **Retrospective boundary check:** RRS profiles computed for five approved antimalarials (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) against the same PfDHFR and corrected PfCRT panels: the framework does not recover the clinically documented resistance signatures (pyrimethamine  $\approx 100\%$  on all four PfDHFR mutants; chloroquine  $\approx 99\%$  on PfCRT K76T/K76A, both within the pipeline-null spread), reinforcing that docking-RRS values are not resistance-circumvention evidence (Supporting Information Table S7).
3. **Physicochemical characterization:** all 17 candidates pass the reported Lipinski/Veber criteria (100% compliance; mean QED 0.703); the available ADMET summary is population-level and is not a candidate-specific safety assessment. Details in Supporting Information Tables S2–S4.
4. **Multi-parameter optimization framework:** a five-component MPO (Vina affinity 35%, DiffDock confidence 25%, QED 20%, ADMET 15%, Lipinski penalty 5%) with a polypharmacology bonus and an MPO  $\geq 0.40$  selection threshold (Section S10).
5. **Methodological rigor:** complete grid box specifications for all four targets, 99.3% cost reduction via centroid-based library reduction, and strict evidence boundaries.

**Evidence boundaries:** the study does not claim experimental affinity, confirmed simultaneous target engagement, biological resistance circumvention, or measured  $\text{IC}_{50}/\text{EC}_{50}$  values. The workflow produces (a) four-target docking profiles from which polypharmacology hypotheses can be generated, (b) per-target RRS values summarizing mutation-panel docking ratios, and (c) exploratory cross-metric associations without conflation.

**Reproducibility and impact:** complete computational provenance (source code, candidate manifests, receptor/ligand files, grid configurations, raw poses, RRS protocols, machine-readable tables) is archived at [https://github.com/NanaEngo/Malaria\\_codesV2](https://github.com/NanaEngo/Malaria_codesV2); the Supporting Information (22 pages, 13 sections, 14 tables, 2 figures) contains the validation datasets and methodological specifications.

The centroid-based reduction strategy lowers the computational barrier for endemic-region groups (94% of malaria cases) by orders of magnitude while retaining structural diversity, providing a reproducible template for hypothesis generation in resistance-challenged therapeutic areas. This manuscript is original, not under consideration elsewhere, and all authors have reviewed and approved the submission. We declare no competing financial interests.

Sincerely,

**Myke Vital Sao Temgoua** (on behalf of all co-authors)